UCS503P - Software Engineering Project

FinShield AI

Intelligent Financial Crime Detection & Investigation Platform

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Chapter 1

Introduction

1.1 Background and Context

Financial fraud and money laundering involve large numbers of transactions that may contain complex and changing patterns. Traditional approaches that depend only on fixed rules may not be sufficient to identify all suspicious activities. FinShield AI is proposed as an intelligent financial crime detection and investigation platform that combines machine learning, rule-based analysis, user behavior analysis, risk scoring, and investigation support.

The system is designed to receive transaction data, validate and preprocess it, analyze the transaction using multiple detection approaches, and assign a risk level. Suspicious transactions can then be converted into alerts and investigation cases for analysts. Explainable AI techniques and an AI-based investigation assistant are planned to support the investigation process.

1.1.1 Problem Statement

Financial institutions must quickly detect suspicious transactions without overwhelming investigators with false positives, a common pitfall of single-method systems. FinShield AI tackles this by unifying machine learning, AML rules, behavioral tracking, and hybrid risk scoring into one platform. Beyond detection, the system delivers clear risk explanations and investigative tools to help analysts act decisively.

1.2 Project Objectives

1. **Objective 1:** Develop a secure platform for managing users, roles, and financial transaction data.
2. **Objective 2:** Detect suspicious financial transactions using machine learning, AML rules, and user behavior analysis.
3. **Objective 3:** Combine different detection results to generate a meaningful risk score and identify high-risk transactions.
4. **Objective 4:** Support fraud investigation through alerts, explainable predictions, case management, and investigation assistance.
5. **Objective 5:** Provide dashboards and reports for analysts, compliance officers, and system administrators.

Chapter 2

Methodology and Design

2.1 System Architecture

FinShield AI follows a modular architecture in which the frontend interacts with the backend through APIs. The backend manages authentication, transactions, fraud analysis, alerts, investigations, and communication with the database. The fraud analysis layer contains machine learning, AML rule processing, and user behavior analysis. Their outputs are intended to be combined by the hybrid risk scoring component.

2.1.1 High-Level Design

The high-level design separates the system into user interaction, application services, fraud intelligence, and data storage components. Customers/users submit transactions through the interface. The backend validates the transaction and passes the required information to the fraud-analysis components. The resulting risk information is stored and can be used by analysts and compliance officers for further action.

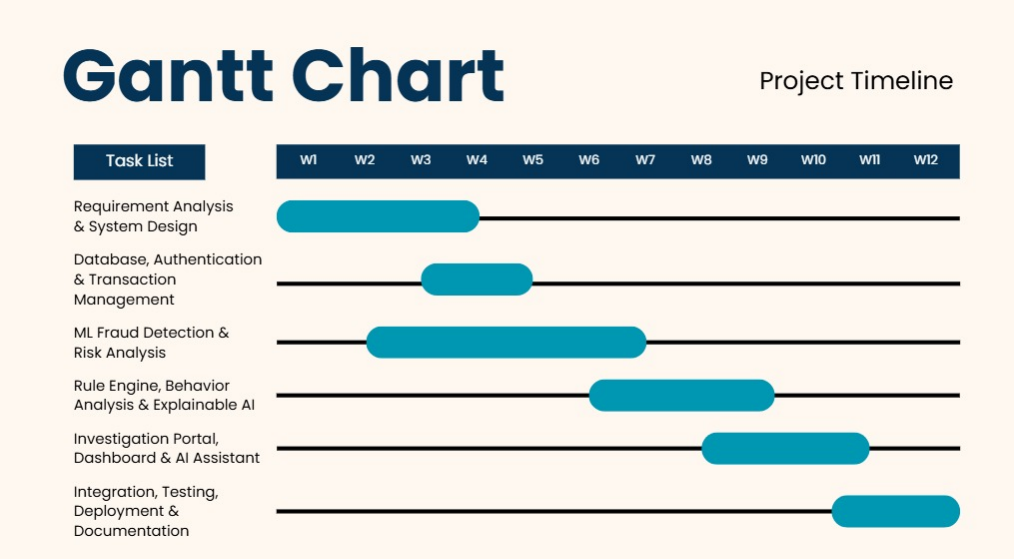


Figure 2.1: Three-Month Project Gantt Chart

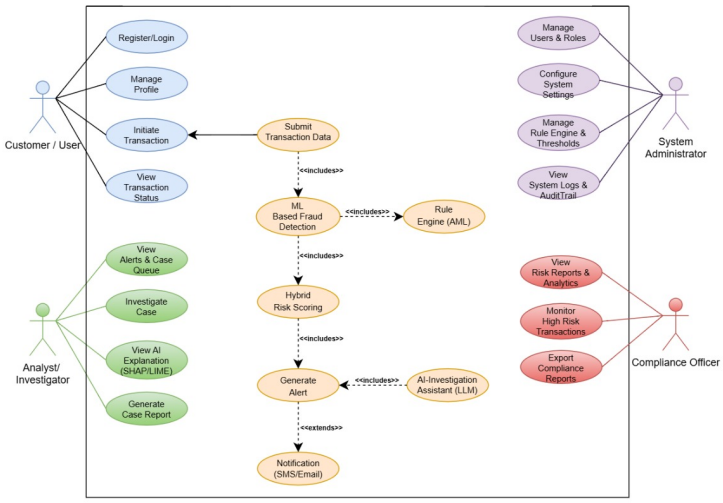


Figure 2.2: Use Case Diagram of FinShield AI

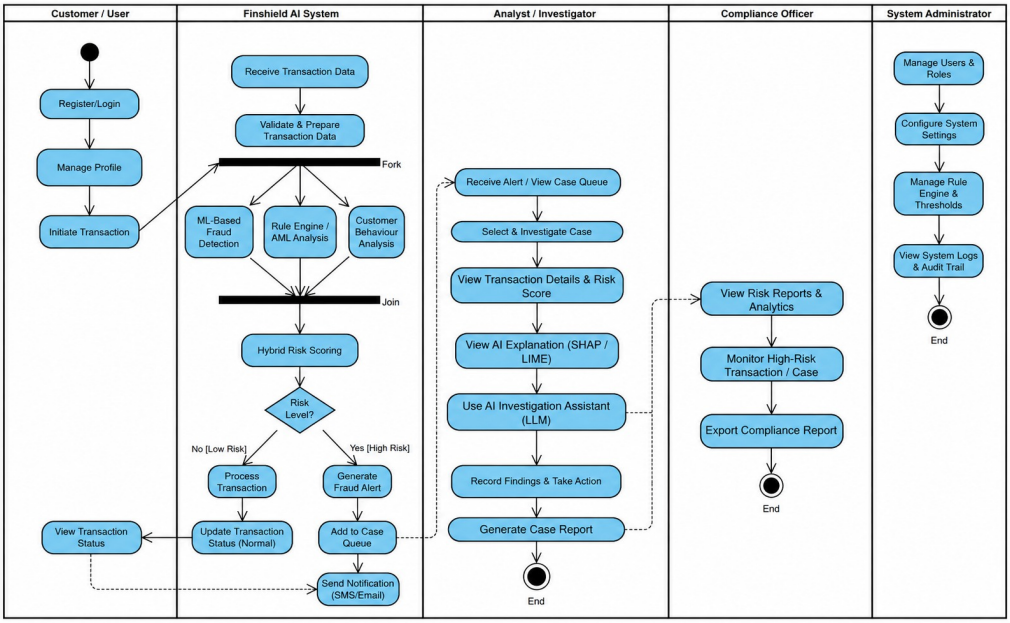


Figure 2.3: Activity Diagram of FinShield AI

UC-01 – User Authentication & Profile Management

1. Use Case Title	User Authentication & Profile Management
2. Abbreviated Title	User Authentication
3. Use Case Id	UC-01
4. Actors	Customer / User
5. Description	This use case allows users to register, log in securely, and manage their profile information.
5.1. Pre Conditions	• User must have a registered account to log in. • System must be available.
5.2. Task Sequence	1. User opens the login/register page. 2. User enters the required information. 3. System validates the information. 4. System authenticates the user. 5. User gets access to the permitted features. 6. User can update profile information.
5.3. Post Conditions	User is successfully logged in. User can access features according to their role. Updated profile information is stored.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.4: Use Case Template: User Authentication & Profile Management

UC-02 – Transaction Management

1. Use Case Title	Transaction Management
2. Abbreviated Title	Manage Transaction
3. Use Case Id	UC-02
4. Actors	Customer / User
5. Description	This use case allows users to initiate financial transactions and allows the system to store and process the transaction information.
5.1. Pre Conditions	• User must be logged in. • Required transaction information must be available.
5.2. Task Sequence	1. User selects the transaction option. 2. User enters transaction details. 3. System validates the information. 4. System stores the transaction. 5. Transaction data is sent for fraud analysis. 6. System updates the transaction status.
5.3. Post Conditions	1. Transaction is stored successfully. 2. Transaction is sent for fraud analysis. 3. User can view the transaction status.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.5: Use Case Template: Transaction Management

UC-03 – ML Fraud Detection & Risk Scoring

1. Use Case Title	ML Fraud Detection & Risk Scoring
2. Abbreviated Title	**Fraud Detection**
3. Use Case Id	UC-03
4. Actors	FinShield AI System
5. Description	This use case analyzes transaction data using machine learning, rules, and user behavior to identify suspicious transactions and calculate their risk score.
5.1. Pre Conditions	• Valid transaction data must be available. • Fraud detection model must be available.
5.2. Task Sequence	1. System receives transaction data. 2. System validates and preprocesses the data. 3. ML model analyses the transaction. 4. Rule Engine checks predefined fraud rules. 5. System analyses user behaviour and transaction patterns. 6. System combines the results. 7. Hybrid risk score is calculated. 8. Transaction is classified according to its risk level.
5.3. Post Conditions	1. Transaction receives a risk score. 2. Transaction is classified as low, medium, or high risk. 3. High-risk transactions are sent for further investigation.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.6: Use Case Template: ML Fraud Detection & Risk Scoring

UC-04 – Fraud Alert & Notification

1. Use Case Title	Fraud Alert & Notification
2. Abbreviated Title	Fraud Alert
3. Use Case Id	UC-04
4. Actors	FinShield AI System, Analyst / Investigator
5. Description	This use case generates an alert when a transaction is identified as high risk and notifies the relevant user.
5.1. Pre Conditions	• Transaction must have been analysed. • Risk score must cross the defined high-risk threshold.
5.2. Task Sequence	1. System checks the transaction risk score. 2. System identifies the transaction as high risk. 3. System generates a fraud alert. 4. Alert is added to the case queue. 5. System sends a notification. 6. Analyst receives the alert.
5.3. Post Conditions	1. Fraud alert is created. 2. Alert is available in the investigation queue. 3. Relevant user is notified.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.7: Use Case Template: Fraud Alert & Notification

UC-05 – Fraud Case Investigation

1. Use Case Title	Fraud Case Investigation
2. Abbreviated Title	Investigate Case
3. Use Case Id	UC-05
4. Actors	Analyst / Investigator
5. Description	This use case allows an analyst to examine suspicious transactions, review risk information, use AI-based assistance, and record investigation findings.
5.1. Pre Conditions	• Analyst must be logged in. • A fraud alert/case must be available.
5.2. Task Sequence	1. Analyst opens the case queue. 2. Analyst selects a suspicious case. 3. System displays transaction details and risk score. 4. Analyst reviews the fraud analysis. 5. Analyst views the SHAP/LIME explanation. 6. Analyst uses the AI Investigation Assistant if required. 7. Analyst records investigation findings. 8. Analyst takes the required action. 9. Analyst generates the case report.
5.3. Post Conditions	1. Case investigation is completed or marked for further review. 2. Investigation findings are stored. 3. Case report is generated.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.8: Use Case Template: Fraud Case Investigation

UC-06 – Explainable AI Analysis

1. Use Case Title	Explainable AI Analysis
2. Abbreviated Title	AI Explanation
3. Use Case Id	UC-06
4. Actors	Analyst / Investigator
5. Description	This use case provides an explanation of why the machine learning model classified a transaction as suspicious using SHAP/LIME techniques.
5.1. Pre Conditions	• Transaction must have an ML prediction. • Analyst must have access to the investigation case.
5.2. Task Sequence	1. Analyst selects a transaction. 2. System loads the ML prediction. 3. System performs SHAP/LIME analysis. 4. System identifies important features. 5. System displays the explanation to the analyst.
5.3. Post Conditions	1. Analyst can understand the main reasons behind the prediction. 2. Explanation is available for investigation.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.9: Use Case Template: Explainable AI Analysis

UC-07 – System & Rule Management

1. Use Case Title	System & Rule Management
2. Abbreviated Title	System Management
3. Use Case Id	UC-07
4. Actors	System Administrator
5. Description	This use case allows the administrator to manage users, roles, system settings, fraud rules, risk thresholds, and system logs.
5.1. Pre Conditions	• Administrator must be logged in. • Administrator must have the required permissions.
5.2. Task Sequence	1. Administrator opens the administration panel. 2. Administrator manages users and roles. 3. Administrator configures system settings. 4. Administrator creates or updates fraud rules. 5. Administrator manages risk thresholds. 6. Administrator views system logs and audit records. 7. System saves the changes.
5.3. Post Conditions	1. System settings are updated. 2. Fraud rules and thresholds are updated. 3. Changes are recorded in the audit trail.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.10: Use Case Template: System & Rule Management

UC-08 – Risk Reports & Compliance Analytics

1. Use Case Title	Risk Reports & Compliance Analytics
2. Abbreviated Title	Risk Reporting
3. Use Case Id	UC-08
4. Actors	Analyst / Investigator
5. Description	This use case allows the compliance officer to monitor high-risk transactions, view fraud analytics, and generate compliance reports.
5.1. Pre Conditions	• Compliance officer must be logged in. • Transaction and risk data must be available.
5.2. Task Sequence	1. Officer opens the analytics dashboard. 2. System displays risk and fraud statistics. 3. Officer views high-risk transactions. 4. Officer analyses risk trends. 5. Officer generates a compliance report. 6. Officer exports the report if required.
5.3. Post Conditions	1. Risk information is reviewed. 2. Compliance report is generated. 3. Report can be exported for further use.
6. Modification History	**Date:** 01-Sep-2026
7. Author	FinShield AI Team

Figure 2.11: Use Case Template: Risk Reports & Compliance Analytics

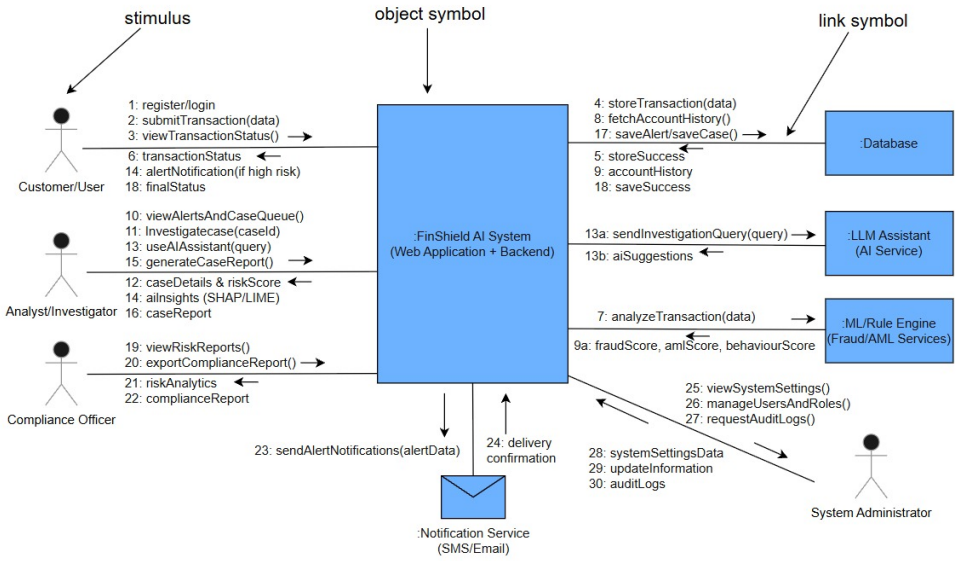


Figure 2.12: Collaboration Diagram of FinShield AI

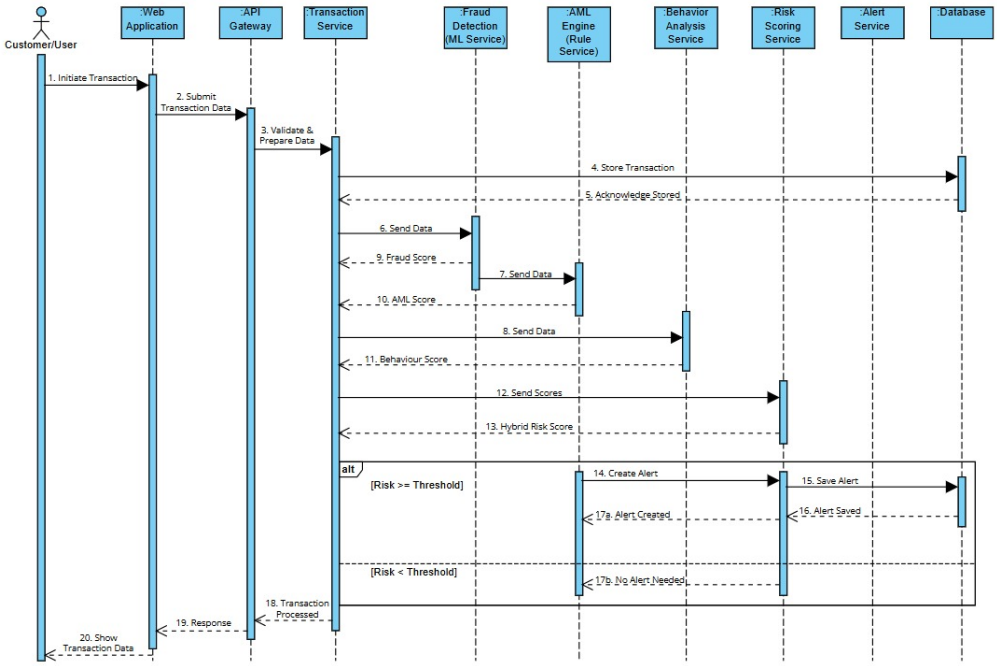


Figure 2.13: Sequence Diagram of FinShield AI

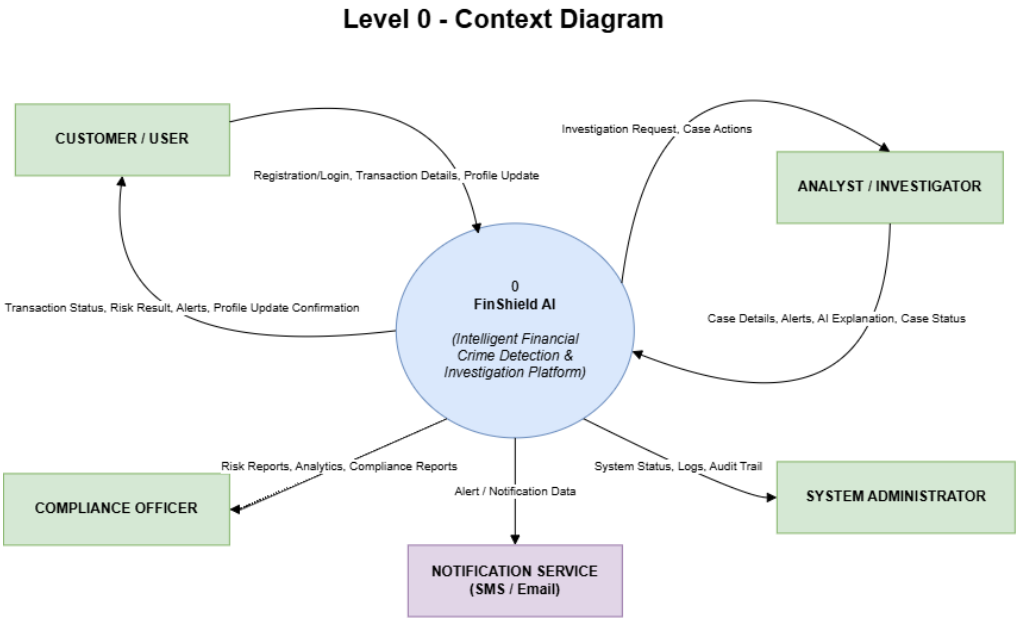


Figure 2.14: Level 0 Data Flow Diagram

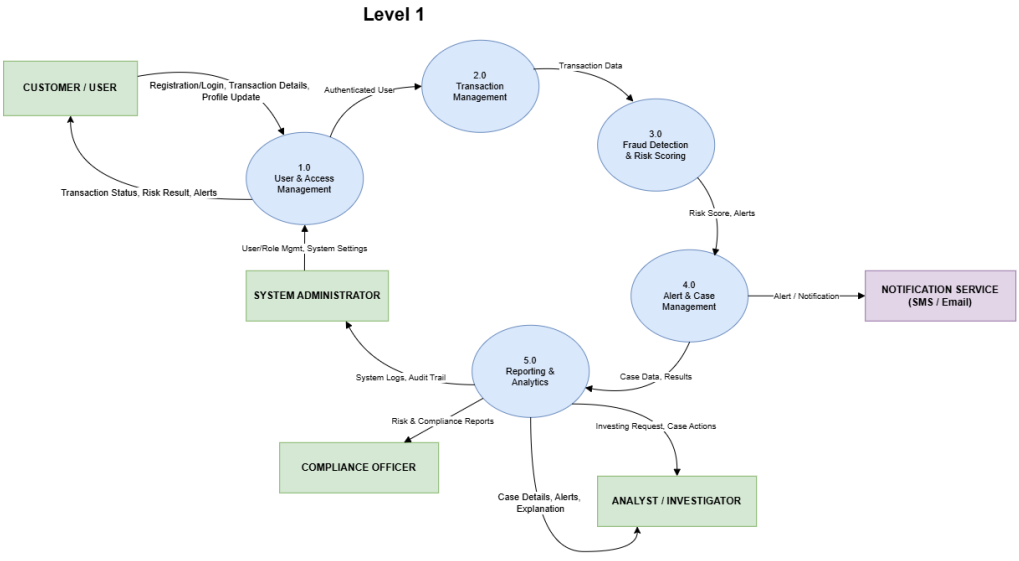


Figure 2.15: Level 1 Data Flow Diagram

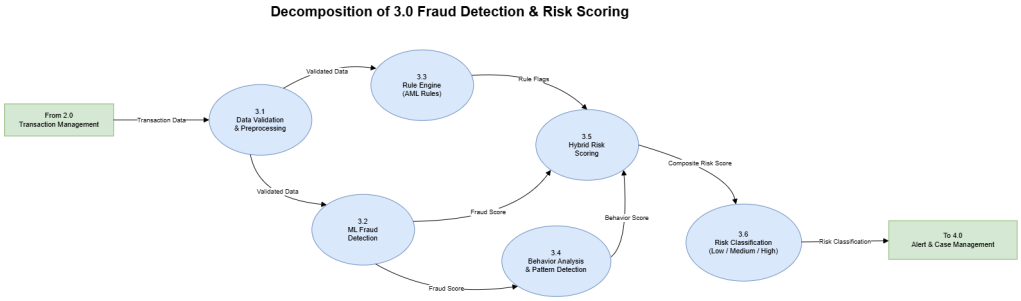


Figure 2.16: Level 2 Data Flow Diagram

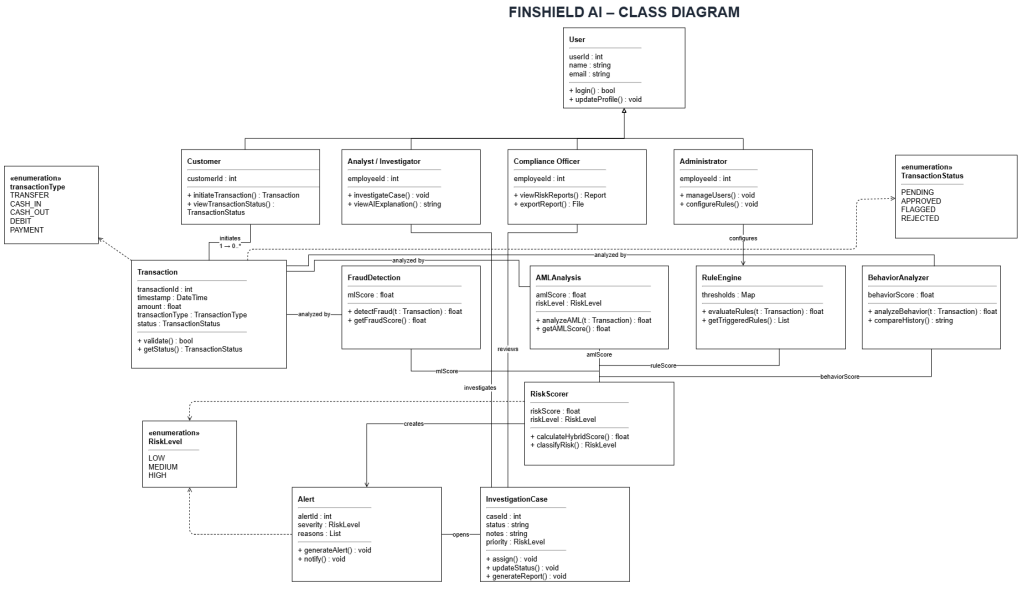


Figure 2.17: Class Diagram

2.1.2 Technical Stack

- **Framework/Language:** Python for machine learning and backend development, with a web-based frontend.
- **Database:** PostgreSQL for structured storage of users, transactions, alerts, investigations, and related records.
- **Tools:** Git and GitHub for version control and collaborative development; Jupyter Notebook for data analysis and model experiments.
- **Libraries:** Pandas and NumPy for data processing, with appropriate machine learning libraries for model development.

2.2 Implementation Details

The prototype implementation is being developed in modules so that each member can work on a specific part while following the same project architecture and common transaction data schema. The initial implementation focuses on the system foundation and the core fraud-detection workflow. The complete investigation, reporting, explainability, and AI-assistant features are planned for later stages of the three-month development timeline.

2.2.1 Prototype Modules Implemented

Authentication and RBAC

The Authentication and Role-Based Access Control (RBAC) module provides secure access to the platform. Users are authenticated before accessing protected functionality, and permissions are controlled according to their assigned roles. The planned roles include Customer/User, Analyst/Investigator, Compliance Officer, and System Administrator. This separation ensures that each actor can access only the functions relevant to their responsibilities.

Transaction Management

The Transaction Management module handles the submission and storage of transaction information. A transaction is first received by the system, validated, and prepared for subsequent fraud analysis. The module provides the basic connection between user transaction input and the fraud-detection pipeline.

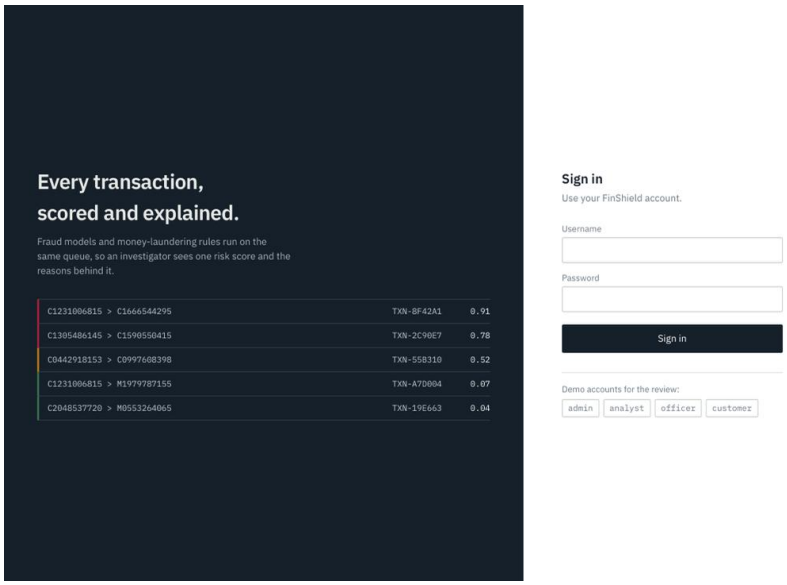


Figure 2.18: Authentication and RBAC

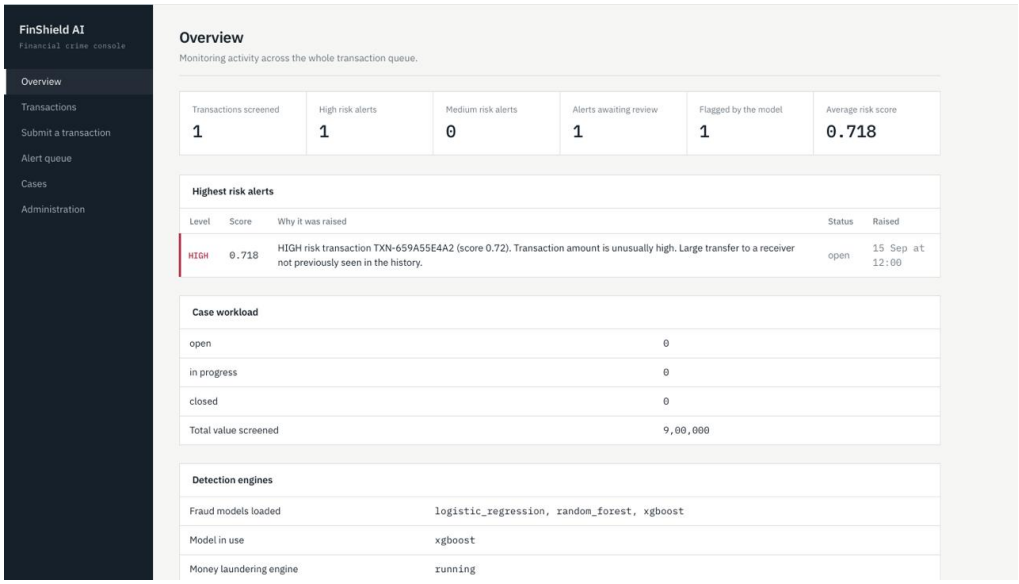


Figure 2.19: Transaction Management

ML Fraud Detection

The ML Fraud Detection module uses transaction attributes to identify potentially fraudulent transactions. The initial development includes dataset analysis, common schema definition, preprocessing, and baseline model development. Model performance will be evaluated using fraud-detection metrics such as precision, recall, F1-score, and ROC-AUC rather than relying only on accuracy.

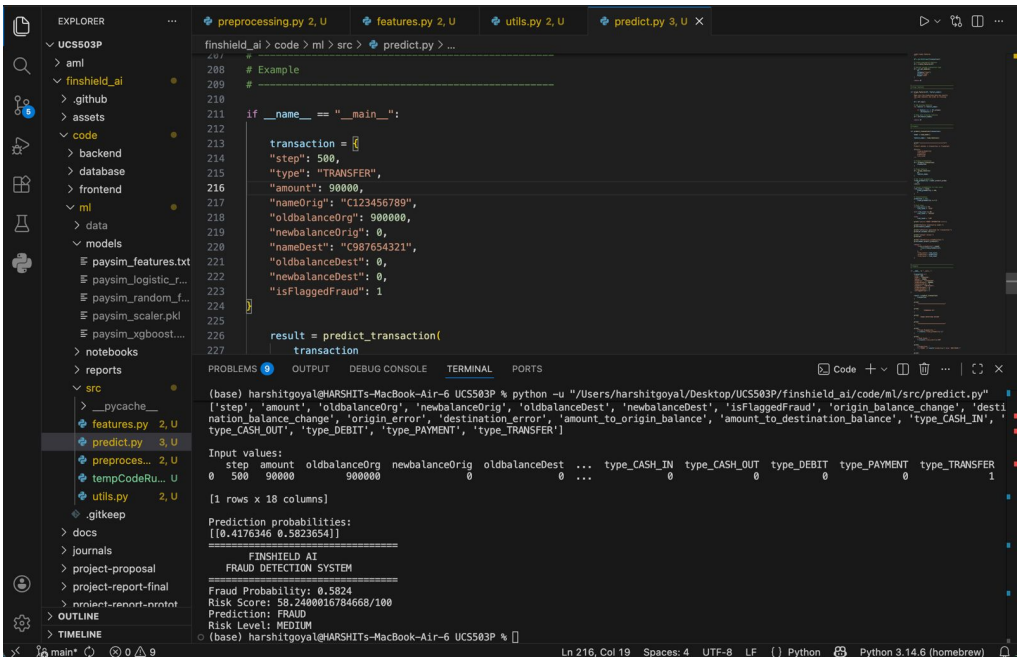


Figure 2.20: ML Fraud Detection

AML Detection

The AML Detection module provides rule-based analysis for identifying transactions that may indicate suspicious financial activity. The module is designed to work with the common transaction schema so that its output can later be combined with the machine learning and behavior-analysis results. The initial stage focuses on defining and testing suitable AML rules; the complete rule engine and hybrid risk-scoring workflow will be developed in later phases.

2.2.2 Initial Data and Schema Preparation

The project uses a common transaction-data schema across all modules. The selected datasets are being studied and prepared so that the ML, AML, backend, and future

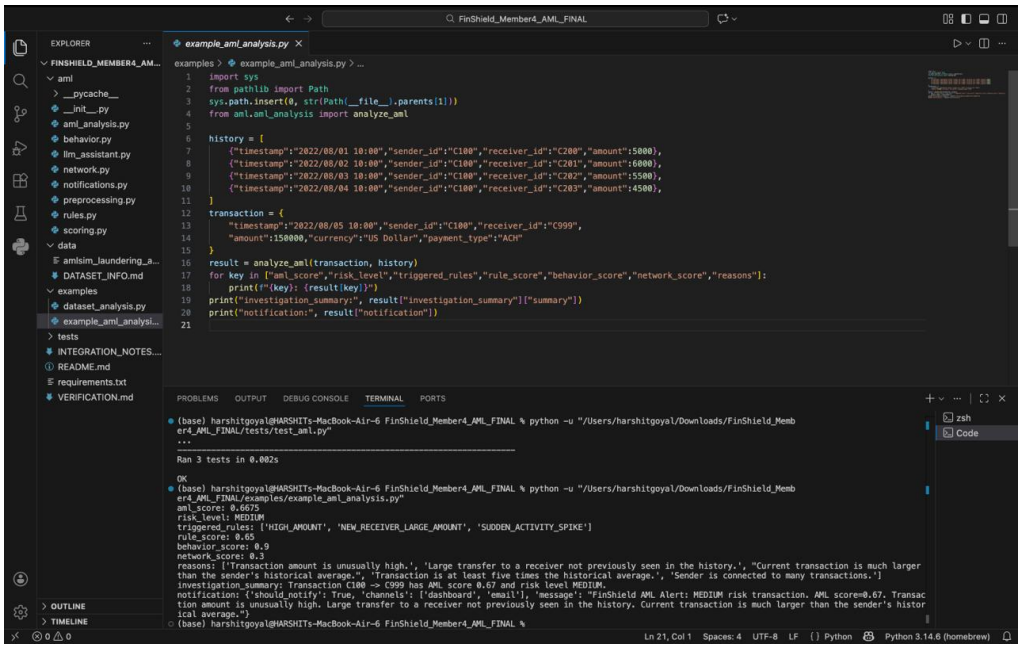


Figure 2.21: AML Detection

behavior-analysis components use a consistent representation of transaction information. A data dictionary is maintained to document the meaning and type of each field. Preprocessing is performed before the data is supplied to the machine learning pipeline.

The schema is treated as a shared interface between team members. Any required change is reviewed by the team before it is introduced, reducing the risk of incompatibility between the ML, AML, backend, and frontend components.

2.2.3 Prototype Development Status

At the prototype stage, the project has moved from requirement analysis and system design toward initial implementation. The current work establishes the foundation for the remaining development phases. The complete system will later integrate behavior analysis, hybrid risk scoring, explainable AI, fraud investigation, dashboards, notifications, compliance reporting, and the AI investigation assistant.